Analysis of gravity in the Randall-Sundrum model

To obtain the degree of Bachelor of Science in Physics

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November 2024

Information

General Information

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Departament	Ciencias Físicas
Study area	High energy physics and gravitation
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Thesis information

Thesis title	Analysis of gravity in the Randall-Sundrum model
Type of thesis	Undergraduate thesis in partial fulfillment of the requirements for the degree of Bachelor of Physics
Abstract	In this work we present a detailed analysis of the Randall-Sundrum model, which proposes a solution to the hierarchy problem in particle physics through the existence of extra dimensions. We have explored the behavior of gauge fields, obtaining the corresponding equations of motion and discussing how these fields interact with the geometry of the extra dimensions. Subsequently, the linearized gravity in this model is computed, focusing on the metric perturbations and the Kaluza-Klein modes, obtaining results consistent with the expected behavior of gravity at low energies on our brane.
Keywords	General Relativity, Alternative theories of gravity, ALAdS

Acknowledgments

This work would not have been possible without the support and teachings of my thesis advisor Rodrigo Aros, to whom I am deeply grateful for proposing and entrusting me with such an interesting topic, and also for all the opportunities he gave me to learn — not only what was necessary for this work, but also for the opportunities that helped me learn more about the world of physics in general.

I also thank the other faculty members of the department, from whom I learned throughout my degree, and especially my classmates who accompanied me along the way, introduced me to the world of gravitation, and helped me understand every concept I could not grasp on my own.

None of this could have happened without the support of my family, especially my father and mother, who have always supported me in everything I have wanted to do. I do not have enough words to thank them for so many years of tireless support and unconditional love.

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Chapter 1

Introduction

In this work, we present a detailed analysis of the foundations and applications of the Randall-Sundrum model, highlighting its relevance in the resolution of open problems in modern physics, such as the hierarchy problem. Particular emphasis is placed on the study of gravity within this model, exploring how its behavior is modified by the geometry of spacetime and evaluating its localization in extra dimensions. Throughout the chapters, we address both the theoretical motivations and recent developments in understanding the relationship between the geometric properties of spacetime and fundamental interactions. This approach aims to bring clarity to the study of extra-dimensional theories, connecting them with the broader efforts to extend the Standard Model toward a more complete and unified framework.

1.1 Standard Model

The development of the Standard Model of particle physics was a collective effort that took decades. Among the contributors we find names such as Dirac, Feynman, Glashow, Weinberg, Salam, Chien-Shiung Wu, Wilczek, and dozens of other theoretical and experimental physicists. In this model, all matter and its interactions are reduced to a small group of fundamental particles. The validity of this model has been repeatedly confirmed experimentally.

Without going into too much detail, for all practical purposes the Standard Model is based on quantum field theory (QFT), and each fundamental particle is described by a field. The interactions are described by a generalization of the 4-vector electromagnetic field given by

$$A^a_\mu \to F^a_{\mu\nu} = \partial_\mu A_\nu - \partial_\mu A^a_\nu + \frac{1}{2} C^a{}_{bc} A^b_\mu A^c_\nu$$

where a, b, c are indices of an internal group, and μ, ν are coordinate indices. $C^a{}_{bc}$ are the

structure constants of the internal group. Analogously, matter is described by Dirac or Majorana fermionic fields Ψ^a.

Some notable models or theories that have shown interesting development over the years include Grand Unified Theories (GUT), supersymmetry, loop quantum gravity, string theory, among others. Within the group of models derived from string theory are those known as *braneworlds*, models in which the current Standard Model of particle physics is sought to be described as the low-energy limit of a larger model.

1.2 Physics Beyond the Standard Model

Despite all its successes, the Standard Model does not present a complete description of the world we know. Most importantly, it includes a description of only three of the four fundamental forces, leaving out gravity. It also presents several inconsistencies or conceptual gaps. For example,

- **Dark energy and dark matter:** If we use only General Relativity as the model of gravity, some of the most interesting observations at the galactic and cosmological level are the apparent lack of matter. This is concentrated, respectively, in the complete failure to describe the rotation of galaxies and the evolution of the universe as a whole. This led some scientists to propose the existence of matter that would not interact with visible matter beyond gravitational interaction. However, this generates a secondary problem, since the Standard Model does not predict the existence of this new matter.

- **Matter-antimatter asymmetry:** The Standard Model cannot fully explain why there is more matter than antimatter in the observable universe.

- **Neutrino masses:** Although the Standard Model includes neutrinos, in its original formulation it considered them to be massless particles. Observations, on the other hand, indicate a small mass. Furthermore, observations also show that neutrinos *oscillate*, that is, they transform into one another, including a change of mass.

- **Unification of fundamental interactions:** The Standard Model describes three of the known fundamental interactions (electromagnetic, weak, and strong) but does not unify them into a single theory. Although the Standard Model unifies the electromagnetic and weak interactions into the electroweak interaction, the strong interaction is described independently through quantum chromodynamics (QCD). Gravity also lies outside the Standard Model.

- **The hierarchy problem:** One of the conceptual pillars of a Grand Unified Theory is the idea that all interactions must merge if the energy at which the interaction occurs is high enough. This is a fact for the electroweak interaction, which results from the unification of electromagnetism and the weak interaction. Under this premise, the large difference between the energy scale of the electroweak interaction (around hundreds of GeV) and the proposed energy scale for gravity (around 10^{19} GeV, known as the Planck scale) is a major problem. In fact, keeping this difference stable without fine-tuning the parameters of the Standard Model is quite problematic.

As is evident, it is necessary to have some concrete proposal to address and resolve some of the aforementioned inconsistencies. With regard to this work, of special interest is the proposal to consider extra dimensions, beyond four, as a possibility to unify apparently distinct theories within a single conceptual framework.

Before proceeding, it is worth mentioning that considering more than four dimensions is far from a novel idea. The first proposal in this direction was made by Theodor Kaluza more than a century ago. By adding an extra dimension to the theory of gravity proposed by Einstein, he was able to formulate an effective theory that unified classical gravity and electromagnetism in five dimensions. In this formulation it is necessary that the fifth dimension be small enough not to be observable without using enormous amounts of energy. It is worth noting that this was proposed even before the creation of quantum mechanics, for which reason the idea of a *small* additional dimension had no particular relevance at the time.

Chapter 2

Elements for the Description of the Model

2.1 Elements of Differential Geometry and General Relativity

General relativity is another of the most successful ideas in physics, in which gravity is described as a deformation of spacetime itself. Its success has been confirmed by multiple observations over the years, and even modified theories of gravity based on these same ideas have emerged. To understand how this theory works, it is necessary to use tools from differential geometry on differentiable manifolds.

A manifold M_n is a generalization of the concept of curves and surfaces to objects of arbitrary dimension. It is a topological space locally homeomorphic to $\mathbb{R}^n$ (which may or may not be different globally), where this local homeomorphism allows us to assign to each point in the manifold a set of coordinates (charts). In the present work we will mostly use the metric formalism of general relativity; however, the vielbein-based formalism is also valid and will be used. This object represents a transformation between the coordinates of the tangent space and the manifold itself, providing an even clearer realization of the equivalence principle, which states that the structure of spacetime in a locally inertial frame has the structure of Minkowski space.

It is possible to take the following coordinates covering a region of our spacetime:

$$x^\mu = \{x^0, x^1, x^2, x^3\} \tag{2.1}$$

Moreover, for each point it is possible to choose a local inertial coordinate system

$$\xi^m = \{\xi^0, \xi^1, \xi^2, \xi^3\} \tag{2.2}$$

whose metric in a neighborhood of the point is Minkowski,

$$ds^2 = \eta_{mn} d\xi^m d\xi^n \tag{2.3}$$

The distance given by our line element in spacetime ds^2 is a scalar, so it is invariant under coordinate transformations. Then, given the local relation between coordinates $d\xi^m$ and dx^μ given by Leibniz's rule,

$$d\xi^m = \frac{\partial \xi^m}{\partial x^\mu} dx^\mu \tag{2.4}$$

the distance in our spacetime takes the following form:

$$ds^2 = g_{\mu\nu} dx^\mu dx^\nu \tag{2.5}$$

The metric $g_{\mu\nu}$ is symmetric and has the following form:

$$g_{\mu\nu} = \eta_{mn} \frac{\partial \xi^m}{\partial x^\mu} \frac{\partial \xi^n}{\partial x^\nu} \tag{2.6}$$

The metric is the element that encodes the geometric properties of spacetime, i.e., the geometric properties of the manifold.

It is also important to define the inverse metric. Let $g^{\mu\nu}$ be the inverse metric; it must satisfy:

$$g_{\lambda\mu} g^{\mu\nu} = \delta^\nu_\lambda \tag{2.7}$$

where the metric tensor and its inverse provide the notion of raising and lowering indices with these objects. An object such as A_μ is called covariant, and an object A^μ contravariant. In general, a tensor can have components of both types and they can be related as follows:

$$\begin{aligned} A_\mu &= g_{\mu\nu} A^\nu \\ A^\mu &= g^{\mu\nu} A_\nu \end{aligned} \tag{2.8}$$

For vector fields in spacetime, the coordinate transformation physically associated with relations between observers is related to the transformation of the infinitesimal coordinate element:

$$x^\mu \to x'^\mu, \tag{2.9}$$

and takes the following form for vector fields:

$$A'^\mu = \frac{\partial x'^\mu}{\partial x^\nu} A^\nu \tag{2.10}$$

In a similar manner, this idea of coordinate transformation can be extended to tensors. A tensor $A^{\kappa\lambda\ldots}_{\mu\nu\ldots}$ is an object which, under a general coordinate transformation, transforms as:

$$A'^{\kappa\lambda\ldots}_{\mu\nu\ldots} = \frac{\partial x'^\kappa}{\partial x^\pi}\frac{\partial x'^\lambda}{\partial x^\rho}\ldots\frac{\partial x^\sigma}{\partial x'^\mu}\frac{\partial x^\tau}{\partial x'^\nu}\ldots A^{\pi\rho\ldots}_{\sigma\tau\ldots} \tag{2.11}$$

Another important definition is that of the covariant derivative. Consider a scalar field $\Phi(x)$; its derivative in spacetime coordinates is

$$\frac{\partial \Phi}{\partial x^\mu} \tag{2.12}$$

which transforms as a tensor.

For vector fields, one can see that the partial derivative with respect to coordinates is not a tensor. Nevertheless, it is possible to see that a 4-vector $\mathbf{A} = e_\mu A^\mu$ is by construction invariant under coordinate transformations and is a scalar, and moreover its partial derivative is a 4-vector:

$$\frac{\partial \mathbf{A}}{\partial x^\nu} = e_\mu \frac{\partial A^\mu}{\partial x^\nu} + \frac{\partial e_\mu}{\partial x^\nu} A^\mu \tag{2.13}$$

The last term represents a measure of the change of tangent vectors at two infinitesimally close points in our spacetime. This comparison is parallel transport and provides the notion of how coordinates change for two locally inertial systems, which is fundamental for comparing physical phenomena under the equivalence principle. It is therefore possible to define this change in the basis vectors e_μ in terms of the connection coefficients $\Gamma^\kappa_{\mu\nu}$, which is not a tensor, as follows:

$$\frac{\partial e_\mu}{\partial x^\nu} = \Gamma^\kappa_{\mu\nu} e_\kappa \tag{2.14}$$

where the connection coefficients show how the basis vectors of the tangent space change when they are parallel-transported between two points of the manifold.

Then, returning to the derivative of the vector field in equation (2.15), we see that it can be rewritten as:

$$\frac{\partial \mathbf{A}}{\partial x^\nu} = e_\kappa \left(\frac{\partial A^\kappa}{\partial x^\nu} + \Gamma^\kappa_{\mu\nu} A^\mu \right) \tag{2.15}$$

where the term inside the parentheses transforms as a tensor, and we define the covariant derivative $D_\nu A^\kappa$ of the 4-vector:

$$D_\nu A^\kappa = \frac{\partial A^\kappa}{\partial x^\nu} + \Gamma^\kappa_{\mu\nu} A^\mu \tag{2.16}$$

From this definition it follows that the covariant derivative acting on a covariant 4-vector takes the following form:

$$D_\nu A_\kappa = \frac{\partial A_\kappa}{\partial x^\nu} - \Gamma^\mu_{\kappa\nu} A_\mu \tag{2.17}$$

Similarly, it is possible to show that the covariant derivative acting on tensors of arbitrary rank takes the form:

$$D_\pi A^{\kappa\lambda...}_{\mu\nu...} = \frac{\partial A^{\kappa\lambda...}_{\mu\nu...}}{\partial x^\pi} + \Gamma^\kappa_{\rho\pi} A^{\rho\lambda...}_{\mu\nu...} + \Gamma^\lambda_{\rho\pi} A^{\kappa\rho...}_{\mu\nu...} + ... - \Gamma^\rho_{\mu\pi} A^{\kappa\lambda...}_{\rho\nu...} - \Gamma^\rho_{\nu\pi} A^{\kappa\lambda...}_{\mu\rho...} \tag{2.18}$$

It is important to note that henceforth we will work on a torsion-free manifold, which provides many important geometric properties in spacetime according to the definitions given above. Furthermore, the connection will specifically be the Christoffel symbols, a connection that is symmetric in its lower indices:

$$\Gamma^\kappa_{\mu\nu} = \Gamma^\kappa_{\nu\mu} \tag{2.19}$$

and moreover, in the absence of torsion, the Christoffel symbols rewritten in terms of the metric take a very characteristic form:

$$\Gamma^\lambda_{\mu\nu} = \frac{1}{2} g^{\lambda\sigma} (\partial_\mu g_{\nu\sigma} + \partial_\nu g_{\mu\sigma} - \partial_\sigma g_{\mu\nu}) \tag{2.20}$$

As mentioned before, the perspective of general relativity is that the gravity we know is a product of the curvature of spacetime; therefore, it is fundamental to understand curvature as a geometric element from the definitions above. We introduce the Riemann tensor, defined from the commutator of covariant derivatives acting on a 4-vector:

$$[D_\kappa, D_\lambda] A_\mu = R_{\kappa\lambda\mu\nu} A^\nu \tag{2.21}$$

which is the (torsion-free) expression for the Riemann curvature tensor, which can also be written in terms of the Christoffel symbols as:

$$R^\rho{}_{\sigma\mu\nu} = \partial_\mu \Gamma^\rho_{\nu\sigma} - \partial_\nu \Gamma^\rho_{\mu\sigma} + \Gamma^\rho_{\mu\lambda} \Gamma^\lambda_{\nu\sigma} - \Gamma^\rho_{\nu\lambda} \Gamma^\lambda_{\mu\sigma}. \tag{2.22}$$

This tensor has two index contractions that give rise to other curvature quantities. The first is a trace of the Riemann tensor, known as the Ricci tensor:

$$R^{\rho}{}_{\sigma\rho\nu} = R_{\sigma\nu} = R_{\nu\sigma} \tag{2.23}$$

where the symmetry of the two indices holds under the condition of zero torsion.

It is also possible to take a full contraction of the Ricci tensor to obtain the so-called Ricci scalar or scalar curvature:

$$R = g^{\sigma\nu} R_{\sigma\nu} \tag{2.24}$$

From these curvature elements of a manifold it is possible to construct the famous Einstein equation (in vacuum), starting by defining the Einstein tensor:

$$G_{\mu\nu} = R_{\mu\nu} - \frac{1}{2} g_{\mu\nu} R + \Lambda g_{\mu\nu} = 0 \tag{2.25}$$

where again the torsion-free condition gives us symmetry under index exchange in the Einstein tensor.

2.2 Kaluza-Klein Theory

In 1919, the German physicist Theodor Kaluza developed a theory that, by introducing a fifth spatial dimension (in comparison to the 4 dimensions of special relativity), unified gravity and electromagnetism, with the latter emerging as a consequence of the extra dimension. Kaluza was unable to give physical, only mathematical, meaning to this extra dimension. In 1926, the Swedish mathematician Oscar Klein reexamined Kaluza's work and made improvements that, in addition to providing a more convincing explanation for why this new dimension was not observed, appeared to have applications in the then-emerging quantum theory.

Kaluza-Klein theory is an excellent framework for showing how to work with compact and additional dimensions beyond the usual spacetime, so understanding it is key not only for its application in this work, but for most theories that involve extra dimensions in the attempt to unify gravity with the other gauge interactions.

The Kaluza-Klein approach to giving meaning to this new dimension is that it behaves differently from the rest of the spacetime dimensions. The extra dimension forms a compact space with a certain compactification scale L; for example, it could be a circle of radius L. For more than one such dimension, it could be a higher-dimensional sphere, a torus, or some other manifold depending on the required compactification. In general, for Kaluza-Klein theories

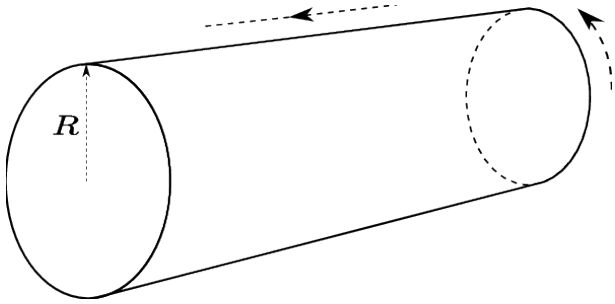

Figure 2.1: Compactification of the extra dimension, image taken from [12]

and their possible extensions, the full D-dimensional spacetime would be given by a direct product $M^4 \times X^{D-4}$, with M^4 the 4-dimensional Minkowski space and X^{D-4} the compact space formed by the extra dimensions (Figure 2.1). This full spacetime must be a solution of Einstein's equations in D dimensions, and it is clear that depending on the scale L, the extra dimensions cannot be observed at large distances given their compact nature.

To observe how these extra dimensions behave, let us analyze scalar fields in this scenario. Considering the indices $A = 0, 1, 2, 3, 5$, where 5 is the index of the extra dimension and we denote the associated coordinate as y, and using the mostly positive metric convention $(- + + + +)$, the action for a massless scalar field is:

$$S = -\frac{1}{2} \int d^5x \partial_A \Phi \partial^A \Phi \tag{2.26}$$

Here the field is a function of all coordinates $\Phi(x^A) = \Phi(x^\mu, y)$. The extra dimension is compactified on a circle S^1 of radius L, so the 5-dimensional spacetime is $M^4 \times S^1$. This compactification of the extra dimension is reflected as a periodicity in the extra coordinate of the fields:

$$\Phi(x^\mu, y) = \Phi(x^\mu, y + 2\pi L) \tag{2.27}$$

Taking a harmonic expansion over the circle, given the periodicity and recalling Fourier modes,

$$\Phi(x^\mu, y) = \Sigma_{n=-\infty}^{+\infty} \phi_n(x^\mu) e^{iny/L} \tag{2.28}$$

note that $\phi_n^* = \phi_{-n}$. Then, the action takes the following form:

$$S = -\frac{1}{2} \int d^4x \int_0^{2\pi R} dy \Sigma_{n,m} \left(\partial_\mu \phi_n \partial^\mu \phi_m - \frac{nm}{L^2} \phi_m \phi_n \right) e^{i(n+m)y/L} \tag{2.29}$$

where one can see that the extra dimension yields a direct integral over the exponential, and using the orthonormality of the mode bases we obtain the following expression:

$$S = -\frac{2\pi L}{2} \int d^4x \Sigma_n \left(\partial_\mu \phi_n \partial^\mu \phi_n^* + \frac{n^2}{L^2} \phi_n \phi_n^* \right) \tag{2.30}$$

We observe that we have an infinite number of 4-dimensional field modes. Introducing a convenient change of variables,

$$\varphi_n = \sqrt{2\pi L}\phi_n \tag{2.31}$$

which allows us to rewrite the action clearly:

$$S = \int d^4x \left[-\frac{1}{2}\partial_\mu\varphi_0\partial^\mu\varphi_0\right] - \int d^4x \Sigma_{k=1}^{\infty}\left(\partial_\mu\varphi_k\partial^\mu\varphi_k^* + \frac{k^2}{L^2}\varphi_k\varphi_k^*\right) \tag{2.32}$$

Here we have separated the zero mode ($n = 0$) from the rest of the infinite sum of modes to better visualize the result. We thus obtain as the complete spectrum of our theory with one compactified dimension:

- A real massless scalar field called the zero mode, φ_0.
- An infinite tower of complex scalar fields with mass inversely proportional to the compactification scale $m_k^2 = k^2/L^2$.

It is important to clarify that, recalling the Klein-Gordon equation, this term is associated with an effective mass given its influence on the dynamics of the fields, and also due to its units, since L has units of distance. Recalling also the mass-energy equivalence and the Planck relation $E = \hbar\nu$, we see that the relation $(1/\text{length})^2$ acts as an effective mass for the dynamics of the fields. Importantly for this result, at low energies (i.e. when $E \ll 1/L$) only the zero mode is relevant; when we are at very high energies ($E \gtrsim 1/L$) all modes are relevant to the behavior of the fields.

Following these ideas, the same treatment can also be applied to abelian gauge fields for the purpose of constructing electromagnetism for theories of this type. Below we will briefly show how the same ideas apply to these fields.

The Maxwell Lagrangian is given by:

$$\mathcal{L} = -\frac{1}{4g_5^2}F_{AB}F^{AB} \tag{2.33}$$

It is possible to show the following useful decomposition of this Lagrangian: $F_{AB}^2 = F_{\mu\nu}^2 + 2(\partial_\mu A_5 - \partial_5 A_\mu)$. Taking again the expansion given the nature of the extra-dimension coordinate y:

$$\begin{aligned} A_\mu(x^\mu, y) &= \Sigma_{n=-\infty}^{+\infty} A_\mu^{(n)}(x^\mu)e^{iny/L} \\ A_5(x^\mu, y) &= \Sigma_{n=-\infty}^{+\infty} A_5^{(n)}(x^\mu)e^{iny/L} \end{aligned} \tag{2.34}$$

Just as for the scalar fields, inserting these expansions into the Lagrangian, building the action, and integrating over the extra dimension—since again an integral over the exponentials in the modes remains—we can obtain the following:

$$S = -\frac{1}{4g_4^2}\int d^4x \left(F^{(0)}_{\mu\nu}F^{(0)\mu\nu} + 2\Sigma_k \left[F^{(k)}_{\mu\nu}F^{*(k)\mu\nu} + \frac{2k^2}{L^2}A^{(k)}_{\mu}A^{*(k)\mu} \right] + 2(\partial_\mu A^{(0)}_5)^2 \right), \quad (2.35)$$

from which again we can see that the spectrum of the 5-dimensional theory with a compact dimension delivers an effective 4-dimensional theory similar to ordinary electromagnetism, except for some contributions that become relevant at very high energies if the compactification scale is small. We thus have in this case:

- A massless gauge field corresponding to the zero mode $A^{(0)}_\mu$ (in the first term).
- A massless scalar field $A^{(0)}_5$.
- An infinite series of modes with mass $\frac{k^2}{L^2}$ identifiable as gauge bosons.

Furthermore, regarding gauge fields, it is important to consider local gauge invariance in 5 dimensions such that $A_M(x^\mu, y) \to A_M(x^\mu, y) + \partial_M \alpha(x^\mu, y)$. After compactification, the 5-dimensional gauge invariance reduces to an infinite number of 4-dimensional transformations for each Kaluza-Klein mode.

Finally, let us see how these ideas work to obtain a unified picture of electromagnetism and gravity in 4 dimensions from the introduction of a compactified fifth dimension. Regarding the compactified dimension, it was Klein who suggested that the compactification scale was of the order of the Planck scale (10^{-35} m) to explain why it had not been directly observed.

Constructing the Einstein-Hilbert action in 5 dimensions, and again considering the compactification of the extra dimension, we have:

$$S = \frac{M_*^3}{2}\int d^5x\sqrt{g_5}R_5 \quad (2.36)$$

In this case g_5 is not the coupling constant from the gauge field case, but rather the determinant of the metric in 5 dimensions. Again, it is possible to expand the metric g_{AB} in harmonic modes since it is a tensor field of this space and satisfies periodicity in the extra dimension.

$$g_{AB}(x^\mu, y) = \Sigma_{n=-\infty}^{+\infty} g^{(n)}_{AB}(x^\mu)e^{iny/L} \quad (2.37)$$

As mentioned before, we aim to show how a unification of the two theories is obtained from the introduction of a new dimension, and since the zero mode is the most relevant in

obtaining the effective theory we seek, we will follow the treatment of [4] and simply show the term in the action corresponding to the zero mode. We will use the Kaluza-Klein metric as an ansatz, defining the following:

$$\begin{aligned} g^{(0)}_{\mu\nu} &= e^{\phi/\sqrt{3}}(g_{\mu\nu}(x) + e^{-\sqrt{3}\phi}A_\mu A_\nu) \\ g^{(0)}_{\mu 5} = g^{(0)}_{5\mu} &= e^{-2\phi/\sqrt{3}}A_\mu \\ g^{(0)}_{55} &= e^{-2\phi/\sqrt{3}} \end{aligned} \tag{2.38}$$

Then, using these elements to build the action expanded in modes and isolating the zero mode from the rest, it is possible to obtain the following expression for the action:

$$S = M_*^3 \pi L \int d^4x \sqrt{g} \left(R_4 - \frac{1}{2}\partial_\mu\phi\partial^\mu\phi - \frac{1}{4}e^{-\sqrt{3}\phi}F^2_{\mu\nu} \right) \tag{2.39}$$

with g the determinant of the 4-dimensional metric. Recalling the Einstein-Hilbert action in 4 dimensions with constants $M^2_{PL}/2$ for gravity, we see it is directly related to our constant, so it is also possible to obtain a relation for Newton's constant in our spacetime with the 4-dimensional one:

$$G_N = \frac{1}{16\pi^2 M_*^3 L} \tag{2.40}$$

And the main result of the analysis, neglecting the possible contributions of higher modes given the compactification scale, is that 4-dimensional gauge fields and 4-dimensional gravity share a common origin from an additional dimension.

The ideas originated by Kaluza and subsequently improved by Klein were of considerable interest; however, their predictions about the contributions of the massive modes were impossible to measure given the compactification scale, rendering them inaccessible. In the early 1950s, there were also attempts to extend this idea with the aim of finding a possible unification in 6 dimensions to incorporate the weak interaction, but they also failed in that potential application, and the ideas of extra dimensions were abandoned until the emergence of string theory.

Chapter 3

The Randall-Sundrum Model

3.1 Model Setup

The Randall-Sundrum (RS) model is a proposal published in 1999 with the objective of providing an explanation for the hierarchy problem in particle physics through a particular mechanism involving a geometry with compact extra dimensions. Subsequently, Lisa Randall and Raman Sundrum found a mechanism with similar characteristics, but without the need for compactification of the extra dimension—a scenario called *large extra dimensions*, whose name indicates that the extra dimension has a scale much larger than the scale usually found in extra-dimension models (mostly the Planck scale).

The RS1 model that we treat in this work assumes the existence of an extra dimension compactified on a circle whose upper and lower halves are identified; that is, we are working on an *orbifold* S^1/Z_2, where S^1 is the one-dimensional sphere (i.e. the circle) and Z_2 is the multiplicative group $\{-1, 1\}$. An *orbifold* is the quotient space produced by a manifold modulo a discrete space group (not just any discrete group). Given these conditions for the extra dimension, whose associated coordinate we will call y, it satisfies an equivalence relation such that $y \sim -y$.

The spacetime of the RS1 model is a 5-dimensional bulk which has, as spatial boundaries along the extra dimension, two 3-branes fixed at the endpoints $y = 0$ and $y = \pi r_c$, where the coordinates on the branes are x^μ, with $\mu = 0, 1, 2, 3$; this is shown in Figure 2.1. An important feature of the model is that, given its proposition as a solution to the hierarchy problem, only gravity can propagate freely across all dimensions, while the other Standard Model fields are confined to one of the branes—which would be the world in which we live—while the other brane would be inaccessible.

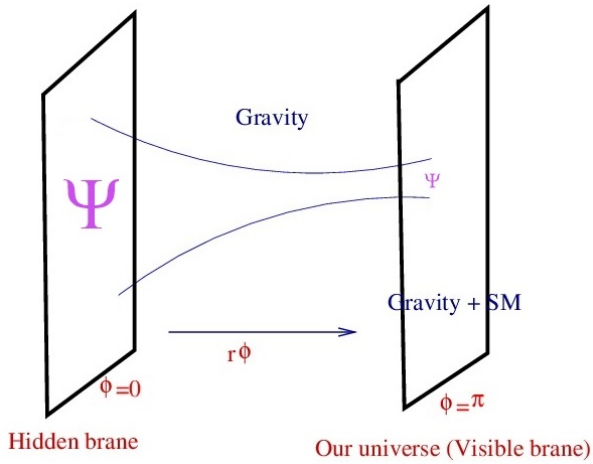

Figure 3.1: Configuration of the RS1 model

The condition of our full spacetime determines certain properties of the extra dimension, just as occurred in Kaluza-Klein theory. Having boundary conditions in our spacetime, these manifest as a periodicity of y in addition to the identification of coordinates (x^μ, y) with $(x^\mu, -y)$, with the range of values of y lying between 0 and π.

Following the proposal of the original publication, the metric of the full 5-dimensional spacetime must be such that it is flat on the boundary branes, since in principle we consider only vacuum energy. Poincaré invariance in the x^μ directions is also required. With this, the non-factorizable metric proposed as an ansatz is a metric containing a *warp factor*—a function that depends only on the extra dimension—acting on the 4-dimensional part of the metric $\eta_{\mu\nu}$, which corresponds to the Minkowski metric in the mostly positive sign convention:

$$ds^2 = e^{-2A(y)}\eta_{\mu\nu}dx^\mu dx^\nu + r_c^2 dy^2 \tag{3.1}$$

and equivalently:

$$g_{MN} = e^{-2A(y)}\eta_{\mu\nu}\delta^\mu_M\delta^\nu_N + r_c^2\delta^5_M\delta^5_N \rightarrow g^{MN} = e^{2A(y)}\eta^{\mu\nu}\delta^M_\mu\delta^N_\nu + r_c^2\delta^M_5\delta^N_5 \tag{3.2}$$

It is important that the *warp factor* be a real function. Then, for this metric—whose explicit form will become apparent at the end of this section—we can construct the action of this model as follows:

$$S = S_{EH} + S_{vis} + S_{hid} \tag{3.3}$$

and each term is explicitly the following:

$$
\begin{aligned}
S_{EH} &= \int d^4x \int dy \sqrt{-g_5}(-\Lambda + 2M_*^3 R) \\
S_{hid} &= \int d^4x \sqrt{-g_{hid}}(\mathcal{L}_{hid} - V_{hid}) \\
S_{vis} &= \int d^4x \sqrt{-g_{vis}}(\mathcal{L}_{vis} - V_{vis})
\end{aligned}
\tag{3.4}
$$

where the first term is the Einstein-Hilbert action for the full space with M_* the 5-dimensional Planck constant, and the other two terms are the contributions from the branes (the visible one and the hidden, inaccessible one for us living on the first). The separation of the matter Lagrangian on the branes from the vacuum energy V, which acts as a gravitational source, is important even if we do not take into account the other fields that might exist on the branes of the model.

3.1.1 T_{MN} on the branes

The 3-branes present in our model have a fundamental contribution as an energy density even in vacuum, providing significant contributions for gravity in the model—the brane tensions. Below, we will show how these tensions work, and subsequently include them in our calculations, since they are a very important element of the model.

If we take the action 3.4 and, based on the principle of least action with the necessary boundary conditions [8], we obtain the equation of motion in 5 dimensions by varying with respect to the inverse metric. This corresponds to the Einstein equation with energy-momentum tensors associated with the brane tensions:

$$
G_{MN} + \Lambda g_{MN} = -(T^0_{MN} + T^\pi_{MN}) \tag{3.5}
$$

If we consider induced coordinates on a general brane as $\sigma^\mu = (\tau, y)$ embedded in a manifold of dimensions larger than the brane, whose coordinates are x^M, it is possible to define the induced metric on the brane by considering reparametrization invariance:

$$
ds^2 = g_{MN} dx^M dx^N = g_{MN} \frac{\partial x^M}{\partial \sigma^\mu} \frac{\partial x^N}{\partial \sigma^\nu} d\sigma^\mu d\sigma^\nu \tag{3.6}
$$

as follows:

$$
h_{\mu\nu} = g_{MN}(x(\sigma)) \frac{\partial x^M}{\partial \sigma^\mu} \frac{\partial x^N}{\partial \sigma^\nu} \tag{3.7}
$$

Following this treatment with the induced metric on the 3-branes of the RS1 model, the energy-momentum tensor of the brane is computed as:

$$
T^{MN} = T \int d^4\sigma \sqrt{\frac{h}{g}} \frac{\partial x^M}{\partial \sigma^\mu} \frac{\partial x^N}{\partial \sigma^\nu} h^{\mu\nu} \delta^{4+n}(x^a - Z^a) \tag{3.8}
$$

Given the positions of the branes in the RS1 model $x^M = (x^\mu, 0)$ and $x^M = (x^\mu, \pi r_c)$, we observe that the terms $\frac{\partial x^M}{\partial \sigma^\mu} = \delta^M_\mu$. In this case, where the coordinates M are those of the full RS1 spacetime, with $y = 0, y = \pi r_c$ for the branes, the right-hand side of equation (3.5) reads:

$$T^{MN} = T_1 \sqrt{\frac{h_{hid}}{g_5}} h^{hid}_{\mu\nu} \delta^\mu_M \delta^\nu_N \delta(y) + T_2 \sqrt{\frac{h_{vis}}{g_5}} h^{vis}_{\mu\nu} \delta^\mu_M \delta^\nu_N \delta(y - \pi r_c) \tag{3.9}$$

Given the following localization of the branes with respect to the full spacetime metric,

$$\begin{aligned} h^{hid}_{\mu\nu} &= g_{MN}(x^\mu, 0) \delta^M_\mu \delta^N_\nu \\ h^{vis}_{\mu\nu} &= g_{MN}(x^\mu, \pi r_c) \delta^M_\mu \delta^N_\nu \end{aligned} \tag{3.10}$$

the induced metrics become evaluated in terms of the *warp factor*:

$$\begin{aligned} h^{hid}_{\mu\nu} &= \eta_{\mu\nu} e^{-2A(0)} \\ h^{vis}_{\mu\nu} &= \eta_{\mu\nu} e^{-2A(\pi r_c)} \end{aligned} \tag{3.11}$$

so that the determinants of the induced metric and the global metric are:

$$\begin{aligned} h^{hid} &= -e^{-8A(0)} \\ h^{vis} &= -e^{-8A(\pi r_c)} \\ g &= -r_c^2 e^{-8A(y)} \end{aligned} \tag{3.12}$$

These results will be useful in the next section when solving the Einstein equations with the vacuum energy density associated with the branes—their tension.

3.1.2 Field equations for the 5 dimensions

Let us now compute the Einstein field equations for the spacetime metric of the model. Recalling from the previous section, the metric was $ds^2 = e^{-2A(y)} \eta_{\mu\nu} dx^\mu dx^\nu + r_c^2 dy^2$, from which we can see that the tensor g_{MN} depends only on the extra dimension, so $\partial_\mu g_{MN} = 0$. With this in mind, we can see that the Christoffel symbols for this case are given by the following expression:

$$\begin{aligned} \Gamma^5_{\mu\nu} &= \frac{1}{2} g^{yR} \left(\partial_\mu g_{R\nu} + \partial_\nu g_{R\mu} - \partial_R g_{\mu\nu} \right) = \frac{1}{2} g^{yR} \left(-\partial_R g_{\mu\nu} \right) = -\frac{1}{2} \eta_{\mu\nu} (-2 e^{-2A} \partial_y A(y)) \\ \Gamma^\mu_{\nu y} &= \frac{1}{2} g^{\mu R} (\partial_y g_{R\nu}) = \frac{1}{2} g^{\mu\lambda} (\partial_y g_{\lambda\nu}) = \frac{1}{2} e^{2A} \eta^{\mu\lambda} \eta_{\lambda\nu} (-2 e^{-2A} \partial_y A) = -\delta^\mu_\nu \partial_y A \end{aligned} \tag{3.13}$$

Now, recalling the definition of the Ricci tensor with the indices of our 5 dimensions,

$$R_{MN} = \partial_P \Gamma^P_{MN} - \partial_N \Gamma^P_{MP} + \Gamma^P_{PQ} \Gamma^Q_{MN} - \Gamma^P_{NQ} \Gamma^Q_{MP} \tag{3.14}$$

and noting directly that $R_{\mu 5} = R_{5\mu} = 0$, we obtain the following for the $R_{\mu\nu}$ and R_{55} components:

$$
\begin{aligned}
R_{\mu\nu} &= \partial_y \Gamma^y_{\mu\nu} + \Gamma^\sigma_{\sigma y}\Gamma^y_{\mu\nu} - \Gamma^\sigma_{\nu y}\Gamma^y_{\mu\sigma} - \Gamma^y_{\nu\sigma}\Gamma^\sigma_{\mu y} \\
&= \partial_y(g_{\mu\nu}\partial_y A) + (-\delta^\sigma_\sigma \partial_y A)(g_{\mu\nu}\partial_y A) \\
&\quad - (-\delta^\sigma_\nu \partial_y A)(g_{\mu\sigma}\partial_y A) - (g_{\nu\lambda}\partial_y A)(-\delta^\lambda_\mu \partial_y A) = ... && (3.15) \\
&= e^{-2A}\eta_{\mu\nu}(-2(\partial_y A)^2 + (\partial_y^2 A)) - 2g_{\mu\nu}(\partial_y A)^2 && (3.16) \\
&= g_{\mu\nu}(\partial_y^2 A - 4(\partial_y A)^2) && (3.17)
\end{aligned}
$$

and for the other component of the Ricci tensor:

$$
\begin{aligned}
R_{yy} &= \partial_S \Gamma^S_{yy} R_{yy} - \partial_y \Gamma^S_{yS} + \Gamma^S_{SL}\Gamma^{yy}_L - \Gamma^S_{yL}\Gamma^L_{yS} && (3.18) \\
&= -\partial_y(-\delta^\sigma_\sigma \partial_y A)(-\delta^\lambda_\sigma \partial_y A) = 4\partial_y^2 A - 4(\partial_y A)^2 && (3.19)
\end{aligned}
$$

Then, constructing the Ricci scalar:

$$
R = g^{MN} R_{MN} = g^{\mu\nu} R_{\mu\nu} + g^{yy} R_{yy} \tag{3.20}
$$

Substituting the Ricci tensors and the metric components directly, we obtain the following:

$$
\begin{aligned}
R &= g^{\mu\nu}[g_{\mu\nu}(\partial_y^2 A - 4(\partial_y A)^2)] + 4(\partial_y^2 A - (\partial_y A)^2) && (3.21) \\
&= 4\partial_y^2 A - 16(\partial_y A)^2 + 4\partial_y^2 A - 4(\partial_y)^2 = 4(2\partial_y^2 A - 5(\partial_y A)^2) && (3.22)
\end{aligned}
$$

With these elements we can construct the Einstein tensor $G_{MN} = G_{\mu\nu} + G_{yy}$:

$$
\begin{aligned}
G_{\mu\nu} &= R_{\mu\nu} - \frac{1}{2} g_{\mu\nu} R && (3.23) \\
&= (g_{\mu\nu}(\partial_y^2 A - 4(\partial_y A)^2)) - \frac{1}{2} g_{\mu\nu}(4(2\partial_y^2 A - 5(\partial_y A)^2)) && (3.24) \\
&= g_{\mu\nu}(3\partial_y^2 A + 6(\partial_y A)^2) = 3g_{\mu\nu}(2(\partial_y A)^2 - \partial_y^2 A) && (3.25)
\end{aligned}
$$

And, for the component of the extra dimension,

$$
\begin{aligned}
G_{yy} = R_{yy} - \frac{1}{2} g_{yy} R = (4(\partial_y^2 A - (\partial_y A)^2) - \\
\frac{1}{2} g_{yy}(4(2\partial_y^2 A - 5(\partial_y A)^2)) = 6(\partial_y A)^2
\end{aligned} \tag{3.26}
$$

Therefore, recalling the model's configuration as shown in [11], and the resulting equation of motion obtained by varying the action with respect to the metric:

$$\sqrt{-g}\, R_{MN} - \frac{1}{2} g_{MN} R = -\frac{1}{4M^3}\Big[\Lambda\sqrt{-g}\, g_{MN} \\ +V_{\text{vis}}\sqrt{-g_{\text{vis}}}\, g^{\text{vis}}_{\mu\nu}\delta^{\mu}_{M}\delta^{\nu}_{N}\delta(\phi-\pi) + V_{\text{hid}}\sqrt{-g_{\text{hid}}}\, g^{\text{hid}}_{\mu\nu}\delta^{\mu}_{M}\delta^{\nu}_{N}\delta(\phi)\Big] \tag{3.27}$$

It should be noted that the cosmological constant in this model is defined for all 5 dimensions, which implies that the G_{55} component of the Einstein equation can provide additional information about the parameters by relating both quantities. Thus, for the extra-dimension component we have:

$$G_{yy} = 6(\partial_y A)^2 = -\frac{\Lambda}{4M_5^3} r_c^2 \tag{3.28}$$

Which admits the following solution for our function $A(y)$:

$$A(y) = \pm\sqrt{\frac{-\Lambda}{24M_5^3}}\, r_c + C \tag{3.29}$$

For simplicity we can define $k = \sqrt{\Lambda/24M_5^3}$. Since the solution has the form $A(y) = \pm Ky$, it is important to recall that the extra dimension has *orbifold* conditions, so invariance under $y \to -y$ is guaranteed.

Now, analyzing the other components of the Einstein tensor, we must take into account the contributions from the brane tensions, evaluating the induced metrics on them and substituting all the data into the equation for those components:

$$\frac{12M_5^3}{r_c^2}(\partial_y^2 A - 2(\partial_y A)^2) - \Lambda = \frac{T_1}{r_c}\delta(y) + \frac{T_2}{r_c}(y-\pi) \tag{3.30}$$

Here we are looking for the value of the brane tensions, so it is useful to consider the form of the equation when y is different from 0 and from π:

$$(\partial_y A)^2 = -\frac{\Lambda}{24M_5^3} r_c^2 + \frac{1}{2}\partial_y^2 A \tag{3.31}$$

Recalling that in equation (3.28) we have information that we can substitute here, we obtain:

$$\frac{12M_5^3}{r_c^2}\partial_y^2 A = T_1\delta(y) + T_2\delta(y-\pi) \tag{3.32}$$

Following the motivation of the original publication [11]—in which the coordinate y of the extra dimension is periodic—the corresponding integration of this equation has some conditions that we look for in the behavior of the *warp factor*. These are listed below, and deeper details of the reasons behind them can be found in [7]:

- Since $e^{-2A(y=0)} = 1$, we have $A(0) = 0$, thus ensuring that the metric on the brane is effectively the Minkowski metric. For this, $C = 0$.

- The integration of equation (3.32) requires that the derivative $\partial_y A$ be discontinuous at the positions of the branes.

Therefore, a function that possesses all these characteristics and remains a solution consistent with the brane tensions we have considered must be a Z_2 symmetry—something akin to a reflection of the extra dimension—so the final solution takes the form:

$$A(y) = \sqrt{\frac{-\Lambda}{24M_5^3}} r_c |y| = k r_c |y|, \tag{3.33}$$

where we can also see that in order to obtain a real solution, the cosmological constant must be negative, so the bulk is an AdS_5 spacetime.

Then, returning to equation (3.32) and integrating, we obtain the following values for the brane tensions:

$$\begin{aligned} T_1 &= 24M_5^3 k \\ T_2 &= -24M_5^3 k \end{aligned} \tag{3.34}$$

Chapter 4

Behavior of Fields in the RS1 Model

4.1 Motivation

In this chapter we present perturbation theory in the Randall-Sundrum model with the aim of understanding how fields are naturally confined to the visible brane given the geometry of the spacetime of the model. We will show the types of solutions that exist for the dynamics of the fields based on their decomposition into Kaluza-Klein modes, and how the zero mode becomes dominant in their analysis.

4.2 Scalar fields ψ

We first analyze the behavior of scalar fields in our spacetime; they will in principle be arbitrary fields that could have dynamical components not only on the branes but also in the bulk.

We consider a free complex scalar field, such that $\psi(x^\mu, y)$; its Lagrangian is given by:

$$\mathcal{L} = \frac{1}{2}\sqrt{-g}\left[g^{MN}\partial_M\psi\partial_N\psi - m^2\psi^2\right] \tag{4.1}$$

In this case the metric g_{MN} is:

$$g_{MN} = \begin{pmatrix} e^{-2k|y|}\eta_{\mu\nu} & 0 \\ 0 & 1 \end{pmatrix} \tag{4.2}$$

so that its determinant is given by:

$$\sqrt{-g} = e^{-4k|y|} \tag{4.3}$$

Then, the explicit form of the Lagrangian is:

$$\mathcal{L} = \frac{1}{2} e^{-4k|y|} \left[e^{2k|y|} \eta^{\mu\nu} \partial_\mu \psi \partial_\nu \psi + (\partial_y \psi)^2 - m^2 \psi^2 \right] \tag{4.4}$$

Considering the 5-dimensional action principle associated with this Lagrangian, the corresponding equation of motion after varying the action with respect to the fields is:

$$\frac{1}{\sqrt{-g}} \partial_M \left(\sqrt{-g} g^{MN} \partial_N \psi \right) - m^2 \psi = 0 \tag{4.5}$$

Substituting our metric explicitly into this equation:

$$\eta^{\mu\nu} \partial_\mu \partial_\nu \psi - e^{4k|y|} \partial_y \left(e^{-4k|y|} \partial_y \psi \right) - m^2 e^{-2k|y|} \psi = 0 \tag{4.6}$$

Making an expansion analogous to the Kaluza-Klein modes for the fields, given the periodicity also present in the extra dimension of the RS1 model:

$$\psi(x^\mu, y) = \Sigma_{n=0}^{\infty} \chi^{(n)}(y) \phi^{(n)}(x^\mu) \tag{4.7}$$

The equation of motion then takes the following form:

$$\sum_n \left[\Box \phi_n \chi_n(y) + \phi_n \chi_n''(y) - 4k \phi_n \,\mathsf{sgn}(y) \chi_n'(y) - m^2 e^{-2k|y|} \phi_n \chi_n(y) \right] = 0. \tag{4.8}$$

where the operator $\Box = \eta^{\mu\nu} \partial_\mu \partial_\nu$, the primes on χ denote derivatives with respect to the coordinate y, and the sign function sgn arises from the derivative of $|y|$. Then, considering this equation of motion for the Kaluza-Klein mode expansion of the scalar fields, we can seek solutions by first analyzing the zero mode:

$$\chi_0''(y) - 4k \,\mathsf{sgn}(y) \chi_0'(y) + m^2 e^{-2k|y|} \chi_0(y) = 0. \tag{4.9}$$

Choosing arbitrary constants in the ansatz solutions, the following is a possible solution for the zero mode:

$$\chi_0(y) = c_1^0 + c_2^0 e^{4ky} \tag{4.10}$$

We can see that this is exactly as expected: a mode that, despite propagating in principle across all dimensions, has its zero-mode propagation (the most relevant at accessible energies) suppressed by a contribution originating from the *warp factor*, so the fields are confined to the brane by this mechanism.

Following the treatment used in [10], it can be shown that the solutions for the Kaluza-Klein modes other than the zero mode have a form known in the literature in terms of Bessel functions:

$$\chi_n(y) = e^{2ky} \left[A_n J_{\pm\sqrt{4+a}} \left(e^{ky} \frac{m_n}{k} \right) + B_n Y_{\pm\sqrt{4+a}} \left(e^{ky} \frac{m_n}{k} \right) \right] \tag{4.11}$$

where A_n and B_n are normalization constants. The reference provides an extensive treatment of these modes, but it is important to note that the behavior of these higher Kaluza-Klein modes is largely determined by their mass. Just as for the zero mode, the contribution of the propagation of these modes along the extra dimension is suppressed by a factor of the form $m_n \to (\ldots)e^{-\pi k r_c}$, so again the mechanism confining the fields to the brane is the *warp factor* and the geometry of spacetime.

4.3 Gauge field A_M

It is possible to apply the same treatment that was used for the scalar fields, this time to gauge fields, in order to study the behavior of electromagnetism in this model.

First, seeking the explicit form of the Maxwell Lagrangian, we construct the following action:

$$S = -\frac{1}{4}\int d^4x \int dy \sqrt{|g|} F^{MN} F_{MN} \tag{4.12}$$

Seeking to preserve the 4-dimensional properties of the tensor F_{MN} in order to recover the observable phenomena when situated on the visible brane, we propose the following form:

$$F_{MN} = \begin{pmatrix} F_{\mu\nu} & F_{y\mu} \\ F_{\mu y} & 0 \end{pmatrix} \tag{4.13}$$

It is possible to show the following decomposition of the tensor F_{MN} (see Appendix 2):

$$F_{MN}F^{MN} = e^{4k|y|}F_{\mu\nu}F^{\mu\nu} + 2e^{2k|y|}F_{\mu y}F^{\mu y} \tag{4.14}$$

Then, using this in equation (4.12) and varying with respect to the fields under the principle of least action, we can obtain equations of motion. Recalling the form:

$$\begin{aligned} F_{\mu\nu} &= \partial_\mu A_\nu - \partial_\nu A_\mu \\ F_{\mu y} &= \partial_\mu A_y - \partial_y A_\mu \end{aligned} \tag{4.15}$$

We then have:

$$\begin{aligned} \delta F_{\mu\nu} &= \partial_\mu(\delta A_\nu) - \partial_\nu(\delta A_\mu) \\ \delta F_{\mu y} &= \partial_\mu(\delta A_y) - \partial_y(\delta A_\mu) \end{aligned} \tag{4.16}$$

We can then see that the term inside the integral takes the following form:

$$\begin{aligned} \delta\left(e^{-4k|y|}F_{\mu\nu}F^{\mu\nu}\right) &= -2e^{-4k|y|}(\partial_\mu F^{\mu\nu})\delta A_\nu \\ \delta\left(2e^{-2k|y|}F_{\mu y}F^{\mu y}\right) &= -4e^{-2k|y|}(\partial_\mu F^{\mu y})\delta A_y - 4e^{-2k|y|}(\partial_y F^{\mu y})\delta A_\mu \end{aligned} \tag{4.17}$$

So the variation of the action from equation (4.12) takes the form:

$$\delta S = \int d^4x \int dy \sqrt{|g|}\left(-\frac{1}{2}e^{-4k|y|}(\partial_\mu F^{\mu\nu})\delta A_\nu - 2e^{-2k|y|}(\partial_\mu F^{\mu y})\delta A_y - 2e^{-2k|y|}(\partial_y F^{\mu y})\delta A_\mu\right) \tag{4.18}$$

Then, integrating by parts we can obtain the following equations of motion for these fields:

$$\begin{aligned} \partial_\mu(e^{-4k|y|}F^{\mu\nu}) + 2\partial_y(e^{-2k|y|}F^{\mu y}) &= 0 \\ \partial_\mu(e^{-2k|y|}F^{\mu y}) &= 0 \end{aligned} \tag{4.19}$$

At this point we can identify that the equation associated with the brane coordinates x^μ is a type of ordinary electromagnetism equation with a *warp* factor, while for the components along the extra dimension we have a more significant modification. In order to preserve all symmetries of the gauge fields, it is necessary to consider the gauge choices $\partial_\mu A^\mu = 0$ and $A_5 = 0$, so that all of electromagnetism, which in principle is free to propagate in the extra dimension, takes the following form:

$$\eta^{\alpha\nu}\eta^{\beta\mu}\partial_\nu F_{\alpha\mu} + \eta^{\beta\gamma}\partial_y(e^{-2ky}\partial_y A_\gamma) = 0 \tag{4.20}$$

Again, we seek a Kaluza-Klein mode expansion of the fields:

$$A_\mu(x^\alpha, y) = \sum_{n=0}^{\infty} A_\mu^{(n)}(x^\alpha) f_n(y) \tag{4.21}$$

In this case, the Kaluza-Klein modes must satisfy the Proca equation, since—as we saw in the previous section—the Kaluza-Klein modes carry a mass factor:

$$\eta^{\mu\nu}\partial_\mu F_{\nu\gamma}^{(n)} = m_n^2 A_\mu^{(n)} \tag{4.22}$$

So the equation of motion for the Kaluza-Klein mode decomposition takes the form of a Sturm-Liouville equation:

$$\partial_y(e^{-2ky}\partial_y f_n) = -m_n^2 f_n \tag{4.23}$$

Turning our attention first to the zero mode, we find for $m_0 = 0$ the following solution:

$$f_0(y) = a_0 + a_1 e^{2ky}, \tag{4.24}$$

where the constants are arbitrary. In particular, a boundary condition requires $f_n' = 0$ at the boundaries, so $f_n = \frac{1}{\sqrt{\pi r_c}}$. We also observe that the zero mode is also localized on the brane.

For the higher Kaluza-Klein modes, returning to the Sturm-Liouville form of the equation, the general form of the wave-function solutions is known in the literature [6]:

$$f_n(y) = e^{ky}\left[A_n J_1\left(e^{ky}\frac{m_n}{k}\right) + B_n J_2\left(e^{ky}\frac{m_n}{k}\right)\right] \tag{4.25}$$

where, if we consider the limit in which r_c is very large, the relation found for the mass factors of the Kaluza-Klein modes is again suppressed by an exponential factor:

$$m_n \to (n + \ldots)\pi k e^{-\pi k r_c} \tag{4.26}$$

4.4 Localized gravity and the hierarchy solution

The last type of field we want to analyze in this work is tensor fields. In particular, we are interested in analyzing the metric and perturbations around that field, since we seek to account for how gravity works in this model.

Recalling that in 5 dimensions:

$$ds^2 = \gamma_{MN} dx^M dx^N \tag{4.27}$$

where the *warp factor* satisfies the following conditions on its derivatives:

$$\begin{aligned} \frac{\partial A}{\partial y} &= -k\,\mathrm{sgn}(y) \\ \frac{\partial^2 A}{\partial y^2} &= -2k(\delta(y) - \delta(y - \pi r_c)) \end{aligned} \tag{4.28}$$

We write the fluctuations around the background metric in the following general form:

$$g_{MN} = \gamma_{MN} + h_{MN} \tag{4.29}$$

Recalling that the action of this model consists of the Einstein-Hilbert term plus the contribution of the two branes, we can use the metric from equation (4.29) to build the E-H action. The natural step is to construct the inverse metric $g^{MN} = \gamma^{MN} - h^{MN}$ and thereby build the Christoffel symbols to first order in the perturbation h. Using the fact that we are not in a flat background in principle—since we are also considering propagation along the extra dimension—we have:

$$\Gamma^A_{BC} = \Gamma^{(0)A}_{BC} + \delta\Gamma^A_{BC}, \tag{4.30}$$

where we have separated the Christoffel symbols associated with the background from those of the perturbation, the latter having the form:

$$\Gamma^{(0)A}_{BC} = \frac{1}{2}\gamma^{AD}\left(\partial_B\gamma_{DC} + \partial_C\gamma_{BD} - \partial_D\gamma_{BC}\right). \tag{4.31}$$

and the perturbation symbols are:

$$\delta\Gamma^{A}_{BC} = \frac{1}{2}\gamma^{AD}\left(\partial_B h_{DC} + \partial_C h_{BD} - \partial_D h_{BC}\right). \tag{4.32}$$

As a direct consequence of this form of the connection, we can show that for a general metric with the structure of the RS1 metric, the Einstein tensor takes the following form by separating the component associated with the perturbation from the background part.

If we consider:

$$\Delta G_{MN} = \delta G_{MN} + k^2 T_{MN}, \tag{4.33}$$

then, separating the background part from the variation (Appendix 3):

$$\delta G_{MN} + k^2\delta T_{MN} = S_{MN} - \frac{1}{2}(h_{MN}R^L_L + g_{MN}g^{LS}\delta R_{LS} - g_{MN}h^{LS}R_{LS}) + k^2\delta T_{MN} \tag{4.34}$$

Regarding the variation of the energy-momentum tensor, we refer to the treatment used in [1] (Appendix 4). It is important to note that, although we are in a vacuum situation in terms of other fields, not only the energy density of the branes contributes in the RS1 scenario, but also the perturbations of the background itself. The perturbed energy-momentum tensor can be obtained directly from the following expression for an arbitrary background:

$$S = \frac{1}{4k^2}\int d^4x\sqrt{-g}h^{\mu\nu}\Delta G_{\mu\nu} \tag{4.35}$$

by varying with respect to the metric $g_{\mu\nu}$, which in the RS1 case would be the induced metric on the branes.

With these elements we can construct the linearization of the curvature tensors to build the E-H action for an arbitrary background, and we can obtain the following expression for the Ricci tensor, again separating the contribution of the background metric from the perturbation, and finally directly constructing the Ricci scalar for the Einstein-Hilbert action. A reference for the detailed treatment of these calculations is [14]; however, I will use only the expression for the Ricci tensor to focus attention on the RS1 model. It is then possible to arrive at the Einstein-Hilbert action for an arbitrary background:

$$\begin{aligned} S = \int d^{4+d}x\frac{1}{4\sqrt{-\gamma}}\left(\nabla_R h_{MN}\nabla^R h^{MN} - \nabla_R h\nabla^R h + 2\nabla_M h_{MN}\nabla^N h - 2\nabla_R h_{MN}\nabla^M h^{RN}\right) \\ + \frac{1}{4}\left(R-\Lambda\right)h_{MN}h^{MN} - \frac{1}{2}hh + G_{MN}h^{MR}R^N - \frac{1}{2}G_{MN}h^{MN}h, \end{aligned} \tag{4.36}$$

where G_{MN} is the Einstein tensor. Then, following the same covariant treatment of [13], it is useful to use the gauge transformation under which this Lagrangian is invariant:

$$h'_{MN}(x,y) = h_{MN}(x,y) - (\nabla_M \xi_N(x,y) + \nabla_N \xi_M(x,y)) \tag{4.37}$$

Then, using this in the action and varying with respect to the background metric, we find the following equation of motion (before contracting the indices):

$$\begin{aligned}
&\frac{1}{2}\nabla_R\nabla^R h_{MN} - \frac{1}{2}\gamma_{MN}\nabla_R\nabla^R h + \frac{1}{2}\nabla_M\nabla_N h + \frac{1}{2}\gamma_{MN}\nabla_R\nabla^S h^{RS} - \\
&\frac{1}{2}\nabla_M\nabla_R h^{RN} - \frac{1}{2}\nabla_N\nabla_R h^{RM} + \frac{k}{2}(h_{MN} + \gamma_{MN}h) + [-4kh_{MN} - k(h_{M\nu}\delta^\nu_N + h_{\mu N}\delta^\mu_M) \\
&+k\gamma_{MN}\tilde{h} + k\delta^\mu_M\delta^\nu_N\gamma_{\mu\nu}h + 6k\delta^\mu_M\delta^\nu_N(\tilde{h}_{\mu\nu} - \frac{1}{2}\gamma_{\mu\nu}\tilde{h})\Big]\,\tilde{\delta} = 0,
\end{aligned} \tag{4.38}$$

where elements of the type $\tilde{h}$ denote the full index contraction of that object (the trace and the determinant for metrics). After contracting the indices, a much more compact equation remains:

$$\frac{3}{2}\nabla_R\nabla^S h^{RS} - \frac{3}{2}\nabla_R\nabla^R h + 6k\frac{1}{2}h - 3k\tilde{h}\tilde{\delta} = 0 \tag{4.39}$$

Then, recalling the contributions of the brane tensions, we must consider how they couple to gravitational perturbations, which has a general form given in [2]:

$$\frac{\kappa}{2}B_1\int h_{\mu\nu}(x,0)T_1^{\mu\nu}\,dx + \hat{\kappa}\frac{1}{2}B_2\int h_{\mu\nu}(x,R)T_2^{\mu\nu}\sqrt{-\gamma_2}\,dx \tag{4.40}$$

This expression is for a general contribution on the branes, but we seek to focus on the behavior of the vacuum solution. If we first consider $T_{\mu\nu} = 0$, the equations of motion separated by component for $h_{\mu\nu}$ and $h_{\mu y}$ are:

For the $\mu\nu$ components:

$$\partial_y\left(\partial_\mu\tilde{h} - \partial_\nu h_{\mu\nu}\right) - 3\partial_\sigma\partial_y\partial_\mu\varphi = 0 \tag{4.41}$$

and for the μy components:

$$\frac{1}{2}\left(\partial_\mu\partial_\nu h_{\mu\nu} - \partial_\mu\partial^\mu\tilde{h}\right) - \frac{3}{2}\partial_\sigma\partial_y\frac{\partial\tilde{h}}{\partial y} + 6k^2\varphi = 0 \tag{4.42}$$

where $A = A(y)$ is the exponent in the RS1 metric and $\phi = h_{yy}(x^\mu)$ is the so-called unitary gauge. From the decomposition of the perturbation h_{MN} we can identify the following degrees

of freedom after moving the brane tension contributions to the other side of the equations of motion:

$$h_{\mu\nu} = b_{\mu\nu} + \gamma_{\mu\nu}\left(A - \frac{2kr_c}{(e^{2kr_c}-1)}\right)\varphi + \frac{1}{2k^2}\left(A - \frac{2kr_c}{(e^{2kr_c}-1)} + \frac{1}{2} + \frac{(e^{2kr_c}-1)}{2kr_c}e^{-2A}\right)\partial_\mu\partial_\nu\varphi \tag{4.43}$$

In this decomposition, for $T_{\mu\nu} = 0$, we have a tensorial degree of freedom $b_{\mu\nu}$ associated with a massless graviton and a massless scalar field φ.

To compute the behavior of gravity in the scenario with the brane tensions involved in the equation of motion, let us examine the propagation of perturbations along the $\mu\nu$ components:

$$\begin{aligned} \frac{\kappa^2}{2}T_{\mu\nu} &= \frac{1}{2M^3} - \Lambda - \lambda_1 e^A\delta(z) - \lambda_2 e^A\delta(z-L_z)g_{\mu\nu} \\ &= \frac{1}{2M^3} - \Lambda e^{-2A} - \lambda_1 e^{-A}\delta(z) - \lambda_2 e^{-A}\delta(z-L_z)(\eta_{\mu\nu} + h_{\mu\nu}) \end{aligned} \tag{4.44}$$

Here we have also moved the cosmological constant to the energy-momentum tensor side. Then, taking the explicit form of (4.36) for the RS1 metric, we have the following expression for the linearization of the Ricci tensor:

$$R_{MN} = \frac{1}{2}\left(\partial_\rho\partial^\rho h_{MN} - \partial_\mu\partial_\nu h^{\mu\nu} + \eta^{\rho\sigma}\partial_\sigma\partial_M h_{N\rho} + \eta^{\rho\sigma}\partial_\sigma\partial_N h_{M\rho}\right) \tag{4.45}$$

Again, the equation of motion obtained by varying the linearized E-H action to first order takes the following form:

$$\kappa^2 T_{MN} = \frac{1}{2}M^3\left(\partial_\mu\partial_\nu h^{\mu\nu} - \partial_\mu\partial^\mu\tilde{h} + 6k^2\varphi\right), \tag{4.46}$$

which, substituting T_{MN}, gives:

$$\left(\Box - \partial_y^2 + 6k^2\right)h_{\mu\nu} = \kappa^2\left(\lambda_1\delta(y) + \lambda_2\delta(y-\pi r_c)\right)\gamma_{\mu\nu} \tag{4.47}$$

One way to solve this equation is to perform a change of variables in the metric perturbation $h_{\mu\nu} \longrightarrow e^{\alpha A(y)}h_{\mu\nu}$, which yields the following form of the equation (without manipulating the right-hand side yet):

$$-\frac{1}{2}\partial_R\partial_R h_{\mu\nu} + \frac{3}{2}\alpha A'(y)h'_{\mu\nu} + \frac{3}{2}\alpha - \frac{1}{2}\alpha^2 A'(y)^2 - \frac{1}{2}\alpha A''(y)h_{\mu\nu} = k^2 T_{\mu\nu}. \tag{4.48}$$

Then, choosing the coefficient $\alpha = 3/2$ and subsequently applying a Kaluza-Klein decomposition to the perturbations:

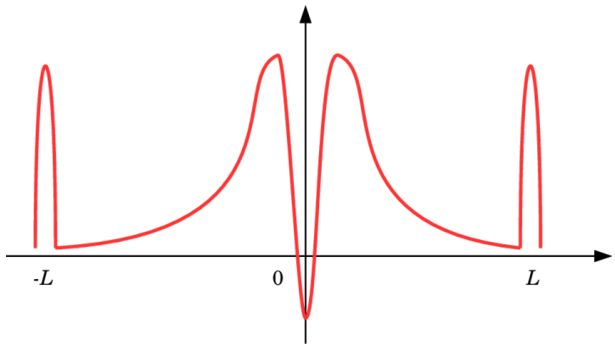

Figure 4.1: Volcano-type potential

$$h_{\mu\nu}(x,y) = \sum_{n=0}^{\infty} h_{\mu\nu}^{(n)}(x)\psi_n(y). \tag{4.49}$$

Considering that the Kaluza-Klein modes have an associated mass given by $\Box h_{\mu\nu}^{(n)} \equiv \partial_\rho \partial^\rho h_{\mu\nu}^{(n)} = m_n^2 h_{\mu\nu}^{(n)}$.

Then, substituting this into the equation for the perturbations and rewriting the contributions from the brane tensions explicitly, we have the following equation:

$$-\psi_n''(y) + \left[\frac{15}{4}k^2\left(k|y|+1\right)^2 - 3k\delta(y) - \delta(y-\pi r_c)\frac{k|y|+1}{k|y|+1}.\right]\psi_n(y) = m_n^2\psi_n(y), \tag{4.50}$$

which can be identified, in very close analogy, with a Schrödinger equation with a volcano-type potential.

We need boundary conditions on the branes for these wave functions. Integrating equation (4.50) over domains close to the branes, we find:

$$-\psi_n'(0^+) + \psi_n'(0^-) - 3k\psi_n(0) = 0. \tag{4.51}$$

Therefore, in order to preserve the characteristic symmetries of spacetime on the *orbifold*, we have:

$$\begin{aligned} \text{for the hidden brane: } \psi_n'(0) &= -\frac{3k}{2}\psi_n(0). \\ \text{for the visible brane: } \psi_n'(L_z) &= -\frac{3k}{2(k\pi r_c+1)}\psi_n(\pi r_c). \end{aligned} \tag{4.52}$$

Furthermore, isolating the zero mode from equation (4.50) (and recalling the relations between the *warp factor* and the brane tensions obtained in (3.33)), we observe:

$$\psi_0'' + \frac{9}{4}A'(z)^2 - \frac{3}{2}A''(z)\psi_0 = 0. \tag{4.53}$$

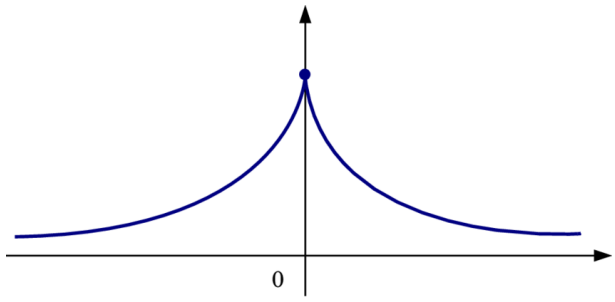

Figure 4.2: Behavior of the zero mode, [5]

We propose an ansatz satisfying the boundary conditions:

$$\psi_0(y) = (k|y| + 1)^{-\frac{3}{2}} \tag{4.54}$$

Taking $A(y) \to A(z)$ as the change of variables shown in the treatment of [5]:
The derivatives of this function are:

$$\begin{aligned} A'(z) &= \frac{k}{k|z| + 1} \\ A''(z) &= -\frac{k^2}{(k|z| + 1)^2}, \end{aligned} \tag{4.55}$$

which, upon substitution into the equation of motion for the zero mode, shows that it is identically zero, since the terms on the left-hand side cancel each other. Thus, the zero mode $\psi_0(y) = (k|y| + 1)^{-\frac{3}{2}}$ has a very particular behavior since its strongest contribution is on the hidden brane (see Figure 3.2). This means that the most important mode at measurable scales is highly suppressed on its way to the visible brane, even for a perturbation originating on our own brane.

Chapter 5

Solution to the Hierarchy Problem for Gravity

5.1 Physical Implications

The RS1 model considers a compactification of the extra dimension that is very small, so it cannot be observed by current experiments or in the near future. For this reason, we seek to understand the relationship between the gravity of the full model—which in principle we cannot observe—and the gravity that is effectively present in our observations.

Returning to the implications of the gravitational perturbations, we consider the form of the perturbations on the visible brane:

$$ds^2 = e^{-2kr_c|\phi|}\left[\eta_{\mu\nu} + h_{\mu\nu}(x)\right] dx^\mu dx^\nu + r_c^2(x) d\phi^2. \tag{5.1}$$

where only the purely 4-dimensional part is perturbed.

Considering the Einstein-Hilbert action mentioned previously, but this time without a cosmological constant:

$$S = 2M_5^3 \int d^5x \sqrt{-\tilde{g}_5} R_5 \subset S \tag{5.2}$$

where S is the action that contains the brane tension terms, the cosmological constant, and in principle any contribution to the energy-momentum tensor from the fields. From the metric (3.2) we can see that the relation between the perturbed ($\tilde{g}$) purely 4-dimensional determinants on the metric and the full metric determinant is:

$$\sqrt{\tilde{g}_5} = r_c e^{-4kr_c|y|} \sqrt{\tilde{g}_4} \tag{5.3}$$

Taking the relation detailed in [3], we see that the relation between the 5-dimensional curvature tensors and the purely 4-dimensional part is:

$$R^{(5d)} = e^{2A(y)} R^{(4d)} + \frac{1}{r_c}(8A'' - 20A'^2) \tag{5.4}$$

Thus, we can see how the purely 4-dimensional part of the E-H action is related to the 5-dimensional one in this way:

$$S = 2M_5^3 \int_{-\pi}^{\pi} dy\, e^{-2kr_c|y|} \int d^4x \sqrt{\tilde{g}_4} R^{(4d)} \tag{5.5}$$

We must also consider the purely brane contribution to the E-H action, since what preceded was the bulk part projected onto the branes. Thus, for the branes we have:

$$S = 2M_4^2 \int d^4x \sqrt{\tilde{g}_4} R^{(4d)} \tag{5.6}$$

Then, the total effective action over the internal brane coordinates σ^μ satisfies:

$$\begin{aligned} S_{eff} &= 4M_5^3 \int_{-\pi}^{\pi} dy\, e^{-2kr_c|y|} \int d^4x \sqrt{\tilde{g}_4} R^{(4d)} \\ &= 2M_5^3 \int d^4x \sqrt{\tilde{g}_4} R^{(4d)} \frac{1}{k}(1 - e^{-2kr_c\pi}) \end{aligned} \tag{5.7}$$

from which we obtain a direct relation for the fundamental scales of both theories—the Planck scale:

$$M_4^2 = \frac{M_5^3}{k}(1 - e^{-2kr_c\pi}) \tag{5.8}$$

Here we see that the mass contributions are affected by the *warp factor*, generating a mechanism that produces an exponential change between the fundamental effective scale on the visible brane. For example, if the *warp factor* is of order $\sim 10^{15}$, this mechanism produces scales from the fundamental mass parameter not far from 10^{19} GeV. It is possible to take either of the two fundamental scales as the chosen input and derive the other, but with the goal of recovering the scale we observe, it is better to take the 4-dimensional Planck mass at the order we know it, and from it derive the full-model scale, giving a possible value of $kr_c \sim 50$. n

Chapter 6

Conclusions

In this work, a detailed analysis of the Randall-Sundrum model (RS1) has been presented, emphasizing its relevance in modern physics—particularly its ability to address the hierarchy problem and the localization of gravity in extra dimensions. Through this analysis, both the theoretical foundations of the model and the physical implications emerging from its geometric configuration have been explored.

One of the most important results obtained is the verification that the warp factor in the RS1 metric is crucial for the localization of gravity and other fields on the visible brane. This mechanism not only provides a natural solution to the hierarchy problem—explaining the disparity between the gravitational and electroweak scales—but also ensures that gravity at low energies behaves consistently with four-dimensional predictions.

The analysis of metric perturbations showed that the zero mode, associated with the massless graviton, is confined to the visible brane, while the higher Kaluza-Klein modes are exponentially suppressed, thereby guaranteeing the recovery of the expected gravitational behavior. The propagation of scalar and gauge fields in this scenario was also investigated, highlighting how the geometry of spacetime affects their dynamics and localization.

These results reinforce the utility of the RS1 model as a theoretical framework that allows one to address the limitations of the Standard Model and opens new perspectives toward the unification of fundamental interactions.

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Appendix

Orbifolds

An orbifold refers to a quotient space between a manifold and a discrete group. The consequences of this configuration for the geometry of spacetime are that fields compactified in this way can be described by expansions in terms of exponentials and sine and cosine functions. The choice of discrete group for generating an orbifold is restricted to space groups, which are defined as the discrete groups of isometries of $\mathcal{R}^n$, and whose classification is known for $n \leqslant 6$. In one dimension, the quotient spaces $\mathcal{R}/\Gamma$ are considered, where Γ corresponds to the two one-dimensional space groups, Z_2 and D_∞.

In this work, only one-dimensional orbifolds were mentioned; in particular S^1/Z_2, which is also equivalent to $\mathcal{R}/D_\infty$. This space is defined by the following characteristic transformations in $\mathcal{R}$, representing the two generators of the group:

$$\begin{aligned} y &\sim y+1 \\ y &\sim -y \end{aligned} \tag{1}$$

Given the compactification on a circle, it is necessary to define the compactification radius $r_c = \frac{1}{2\pi}$, which naturally sets the scale. This orbifold has two fixed points at $y = 0$ and $y = 1/2$.

It is important to emphasize that we make a clear distinction between the definition of the orbifold, given by equation (1), and any physics defined on it, which is specified by:

$$\begin{aligned} \varphi(-y) &= p\varphi(y) \\ \varphi(y+1) &= q\varphi(y) \end{aligned} \tag{2}$$

This possibility for the fields opens up the option of expanding them in harmonic modes, whose justification is developed much more rigorously in [9].

Explicit form of the Maxwell Lagrangian in the RS1 model

The Maxwell Lagrangian constructed in Section (3.3) takes a particular form when the $\mu\nu$ components are separated from the components involving the coordinate y. We begin by constructing the tensor $F_{MN} = \partial_M A_N - \partial_N A_M$ which, seeking to preserve the properties observed for $F_{\mu\nu}$, takes the following form:

$$F_{MN} = F_{\mu\nu} + F_{\mu y} + F_{y\mu} + \cancel{F_{yy}} \tag{3}$$

Then, recalling the form of the metric (3.2), we can write the following expression for the F^2 term:

$$F_{AB}F^{AB} = g^{AC}g^{BD}F_{AB}F_{CD} \tag{4}$$

Substituting the metric directly:

$$F_{AB}F^{AB} = \left(e^{2A(y)}\eta^{\mu\nu}\delta^A_\mu\delta^C_\nu + \frac{1}{r_c^2}\delta^A_y\delta^C_y\right)\left(e^{2A(y)}\eta^{\lambda\rho}\delta^B_\lambda\delta^D_\rho + \frac{1}{r_c^2}\delta^B_y\delta^D_y\right)F_{AC}F_{BD} \tag{5}$$

Expanding all terms and contracting the indices with the delta functions, without substituting the explicit form of F_{AC}, we obtain:

$$F_{AB}F^{AB} = e^{4A(y)}\eta^{\mu\nu}\eta^{\lambda\rho}F_{\mu\lambda}F_{\nu\rho} + e^{2A(y)}\eta^{\mu\nu}F_{\mu y}F_{\nu y} + e^{2A(y)}\eta^{\lambda\rho}F_{y\lambda}F_{y\rho} \tag{6}$$

From which we directly obtain that the expression for F^2 for this metric, given its form, is:

$$F_{AB}F^{AB} = e^{4A(y)}F_{\mu\nu}F^{\mu\nu} + e^{2A(y)}F^{\mu y}F_{\mu y} \tag{7}$$

With this, the associated action can be directly constructed:

$$S = -\frac{1}{4}\int d^5x F_{MN}F^{MN} \tag{8}$$

which explicitly is:

$$S = -\frac{1}{4}\int dy \int d^4x \left[e^{4A(y)}F_{\mu\nu}F^{\mu\nu} + e^{2A(y)}F^{\mu y}F_{\mu y}\right] \tag{9}$$

General form of the perturbations and gauge invariance

In equation (4.34) we introduced a separation of the background components from the perturbation components, where the background part takes the general form of the curvature tensors associated with an arbitrary metric, while the perturbation contains all the additional

components related to h_{MN}. We begin by considering that the variations of the metric are the perturbations, i.e.:

$$\begin{aligned} \delta g_{MN} &= h_{MN} \\ \delta g^{MN} &= -h^{MN} \end{aligned} \tag{10}$$

The explicit form of the tensor associated with the variation of the Ricci tensor is:

$$S_{MN} \equiv \delta R_{MN} = \frac{1}{2}(\nabla_N \nabla_M h - \nabla_L \nabla_M h^L_N - \nabla_L \nabla_N h^L_M + g^{RL} \nabla_R \nabla_L h_{MN}) \tag{11}$$

where the covariant derivatives are with respect to the background metric.

The explicit form of the covariant derivatives of the perturbation takes the following (rather unfriendly) form:

$$\begin{aligned} \nabla_L \nabla_R h_{MN} = [\partial_R \partial_L - \Gamma^S_{LR} \partial_S] h_{MN} + [\Gamma^S_{LM} \Gamma^K_{RN} + \Gamma^S_{LN} \Gamma^K_{RM}] h_{KS} \\ + [\Gamma^S_{LN} \Gamma^K_{RS} + \Gamma^S_{LR} \Gamma^K_{SN} - \partial_L \Gamma^K_{RN} - \Gamma^K_{RN} \partial_L - \Gamma^K_{LN} \partial_R] h_{KM} \\ + [\Gamma^S_{LM} \Gamma^K_{RS} + \Gamma^S_{LR} \Gamma^K_{SM} - \partial_L \Gamma^K_{RM} - \Gamma^K_{RM} \partial_L - \Gamma^K_{LM} \partial_R] h_{KN} \end{aligned} \tag{12}$$

Using properties of the covariant derivative, we can finally rewrite the tensor associated with the variation of the Ricci tensor:

$$\delta R_{MN} = \frac{1}{2}(\nabla_N \nabla_M h - \nabla_L \nabla_M h^L_N - \nabla_N \nabla_L h^L_M + h^S_N R_{SM} + h^S_M R_{SN} - 2h^{LS} R_{MLNS} + \Box h_{MN}) \tag{13}$$

Up to this point, everything has been done for a general metric, which we then simply substitute with the metric associated with the model.

The equation of motion associated with the perturbation (4.34) has a great advantage when considering its invariance under gauge transformations. In general, for an infinitesimal coordinate transformation $x^M = x^M + \epsilon^M$, an arbitrary metric transforms as $g_{MN} = g_{MN} + \nabla_N \epsilon_M + \nabla_M \epsilon_N$, and it can be shown that for the perturbation h the transformation is analogous to that of the metric. Thus, the existence of this family of perturbations that obey this transformation law and describe the same physical situation gives us freedom to simplify the description by choosing a more convenient reference frame [15]. In the 5-dimensional case of our metric, the 15 independent components of the perturbation have 5 redundancies under this transformation, so it can be described by just 10 independent components. The other gauge choices have been extensively studied in the literature and are also valid in this case, such as the transverse-traceless gauge and the harmonic gauge $h^M_M = 0$, $\partial_M h = 0$. Additionally, for perturbations on the RS1 metric, it is common to use an axial gauge $h_{\mu y} = 0$ so as not to alter too much the structure of the fixed-brane model due to the nature of these gravitational perturbations.

Perturbed energy-momentum tensor δT_{MN}

In the consideration of constructing a perturbation of the tensor T_{MN} in the context of *braneworlds* in general, it is useful to recall the Einstein-Hilbert action:

$$S_{EH} = \frac{1}{2k^2}\int \sqrt{-g}R = \ldots = \frac{1}{2k^2}\int d^4x\sqrt{-g}g^{\mu\nu}(\Gamma^{\alpha}_{\mu\beta}\Gamma^{\beta}_{\nu\alpha} - \Gamma^{\alpha}_{\mu\nu}\Gamma^{\beta}_{\alpha\beta}) + \text{B.T.} \tag{14}$$

Here we have simply rewritten the Ricci scalar and rearranged terms, separating boundary terms through integration by parts. The convenience of this form is that, for purely variational purposes, it is somewhat simpler to obtain the form of the equation of motion for the perturbed metric for an arbitrary background up to second order. Considering the following form of the connection at first order:

$$\Gamma^{\alpha}_{\mu\nu} = \frac{1}{2k^2}g^{\alpha\beta}(\nabla_\nu h_{\beta\mu} + \nabla_\mu h_{\beta\nu} - \nabla_\beta h_{\mu\nu}) \tag{15}$$

It is possible to show that the general form of the second-order expansion is:

$$\begin{aligned} S_{EH} = \frac{1}{2k^2}\int d^4x\sqrt{-g}(g^{\mu\nu}R^{(2)}_{\mu\nu} - h^{\mu\nu}R^{(1)}_{\mu\nu} + h^{\mu}_{\sigma}h^{\sigma\nu}R_{\mu\nu}) \\ -\frac{1}{4k^2}\int d^4x\sqrt{-g}h(g^{\mu\nu}R^{(1)}_{\mu\nu} - h^{\mu\nu}R_{\mu\nu}) \\ -\frac{1}{16k^2}\int d^4x\sqrt{-g}(h^2 - 2h_{\alpha\beta}h^{\alpha\beta})R \end{aligned} \tag{16}$$

where, if we momentarily think of the background as flat space, the second-order action takes the following form in terms of the perturbation:

$$S^{(2)}_{EH} = \frac{1}{8k^2}\int d^4x\, h^{\mu\nu}\left(\partial_\mu\partial_\nu h - \partial_\mu\partial_\alpha h^{\alpha}_{\nu} - \partial_\nu\partial_\alpha h^{\alpha}_{\mu} + \partial_\alpha\partial^\alpha h_{\mu\nu} - \eta_{\mu\nu}\partial_\alpha\partial^\alpha h + \eta_{\mu\nu}\partial_\alpha\partial_\beta h^{\alpha\beta}\right) \tag{17}$$

One can identify that the compact form of the action is:

$$I^{(2)}_{EH} = \frac{1}{4k^2}\int d^4x\, h^{\mu\nu}\delta G_{\mu\nu} \tag{18}$$

whose generalization to an arbitrary background is:

$$I^{(2)}_{EH} = \frac{1}{4k^2}\int d^n x\sqrt{-g}\, h^{of}S_{of} \tag{19}$$

where the lowercase Latin indices are the same as before but are used to mark a distinction for the generalization of the background in this final result, and the tensor S_{of} corresponds to the tensor ΔG_{of} (or ΔG_{MN} for the RS1 indices) from equation (4.33). This result is important because, given the equations we know for the perturbations of the Einstein tensor, we can vary this action with respect to the tensor S_{of}—where this equation is also invariant under the general gauge transformation for h—and obtain an equation for the energy-momentum tensor, since $\Delta G_{MN} = 0$ when varied with respect to h^{MN} (a fully formal treatment of this can be found in [8]).